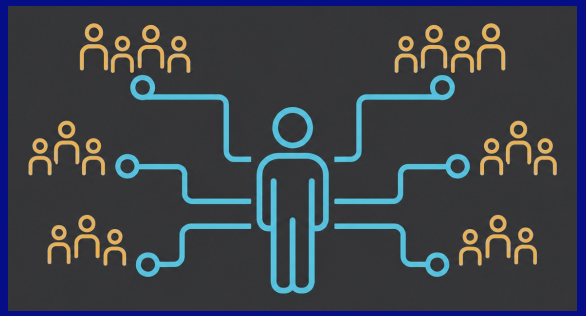


**DHUM101**
**Socially Responsible Computing**

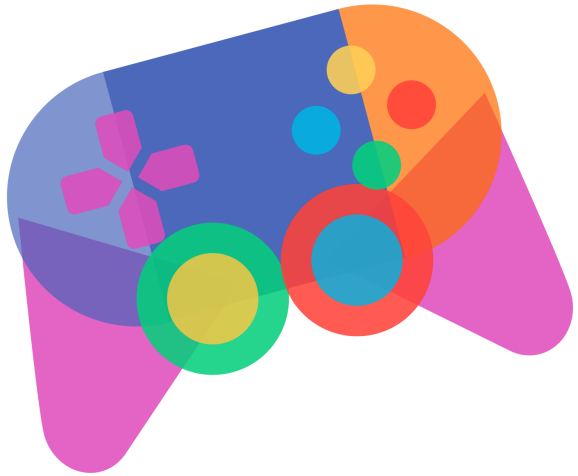
University of Otago
ŌTĀKOU WHAKAIHU WAKA
Te Kura Rorohiko
School of Computing
Te Reo Inarahi me te Mātai Wetereo
English and Linguistics

# Lecture 8
# Identity and Culture

- Stephen Cranefield
  Semester 2, 2026

# OCSS game jam!

september 19th & 20th
9am to 6pm
otago business school G.17



register for free at
ocss.nz/jam

# Updates to Lecture 7

## The data centre backlash

- Video report: Everyone hates datacentres. Do we really need them? (15 min.)
- Report on a planned hyperscale data cente at Auchtertool, Fife, Scotland:
  - Population 635 people (Wikipedia)
  - Proposed details: an area larger than 100 football pitches (larger than the village) and 35 metres high.
- The datacenter backlash is bringing the entire political spectrum together – against big tech billionaires

## Social sustainability

- Report on a settlement of the US federal court case against Meta: Payment of $18bn over a decade, safeguards for teenage users including a daily usage limit of two hours, blocking social media use at night, disabling notifications during school hours and a ban on displaying "likes". But, the payment is "petty cash" to Meta.
- Uber drivers launch European class action over 'soulless' and 'scary' AI algorithm

# The essay assignment

Available now. Due 9am, Thursday 17 September.

Worth 25% of your final mark.

Word Count: 1500 words (up to 10% above this is OK).

**For this essay, consider an emerging digital technology product type. Your job is to consider the technology and assess what ethical issues it raises. First, describe the technology briefly, spending no more than 1/5th of the essay (300 words) on that description. Then, discuss/analyse/report on its potential personal, social, cultural and environmental impacts (good and bad).**

Select a technology from this list.

- EFTPOS via biometrics
- Humanoid household robots
- Fully driverless vehicles
- Body Area Network
- AI traffic control
- AI companions for the elderly
- AI personalised learning at scale
- Brain-computer interface for members of the military/defense
- Schooling via an online 'metaverse'
- AI-controlled cloud-connected household appliances

Consider the following questions or topics (where relevant)

- Privacy
- Autonomy
  - Authenticity
  - Agency
- Social impact
- Sustainability
- Life cycle of development

More details are given in the assignment instructions in Aoroa, including the rubric we will use to mark submissions.

# Computing and user identity

# From the Guiding Questions for Ethics of Emerging ICT (Lecture 4)

"Key ethical questions refer to personal identity. … By taking on and enhancing traditionally human functions, emerging technologies may alter our view on what it means to be a human or individual. This includes conceptions of authenticity, human dignity, normality,  and the idea of what makes someone healthy."

- Guiding questions related to Identity:
  - Does the Emerging Technology change human capabilities, e.g. their ability to perform certain tasks? (Discussed in Tutorial 7)
  - How will the Emerging Technology affect the way in which users see themselves or one another? (Discussed in this lecture)

*Reference:* Stahl, Timmermans and Flick, "Ethics of Emerging Information and Communication Technologies: On the implementation of responsible research and innovation", 2017. https://doi.org/10.1093/scipol/scw069

# Identity is complicated

Some definitions:

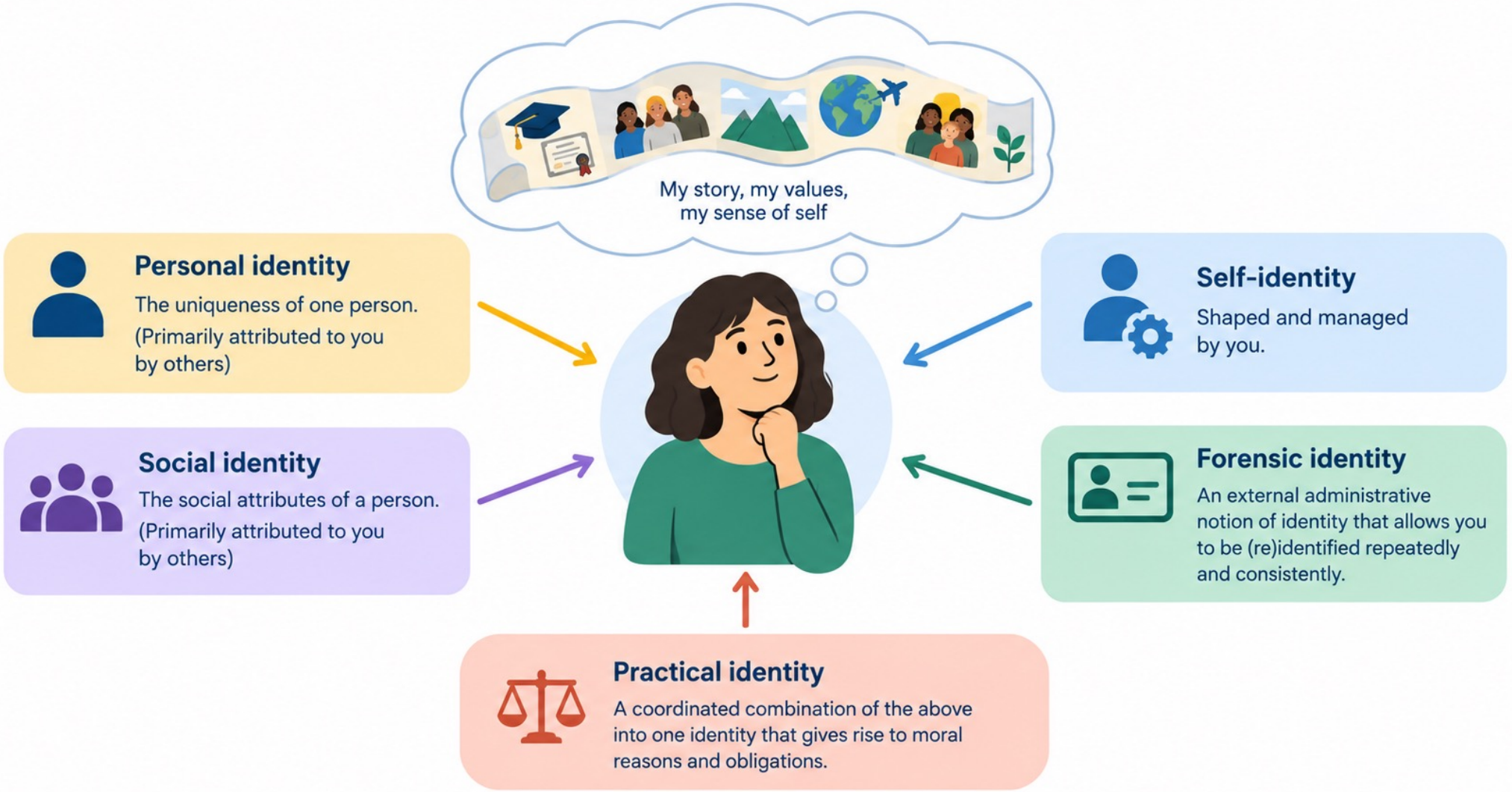
- **Personal identity**: the uniqueness of one person.
- **Social identity**: the social attributes of a person.

} Primarily attributed by others. May arise even before you are born and continue after you are dead.

- **Self-identity**: shaped and managed by you. You can experiment with and change this despite the normative expectations of society.
- **Forensic identity:** an external administrative notion of identity that allows you to be (re)identified repeatedly and consistently.
- **Practical identity**: A coordinated combination of the above into one identity that gives rise to moral reasons and obligations.
  - One view is that you maintain an autobiographical *narrative* "embedded in a social context with memories, contacts, role-playing and emotions". It allows you to make sense of yourself.

*Reference:* Noëmi Manders-Huits, "Practical versus moral identities in identity management", 2010. https://link.springer.com/article/10.1007/s10676-010-9216-8

# Identity is complicated
## (ChatGPT's version)



# The algorithmic self

- **Algorithmic Self**: "a form of digitally mediated identity in which personal awareness, preferences, and even emotional patterns are shaped through continuous feedback from AI systems. … It is not merely a self reflected in technology but co-constructed by it — where algorithms do not passively reflect the self but actively participate in its formation".
- Not just about direct use of AI. Also relevant to algorithmically generated information, recommendations, jobs, ratings, etc. on social media, gig work platforms and so on.
- Some issues:
  - **Algorithmic identity shaping**, e.g. the annual Spotify Wrapped; A study showed that "many accepted the summaries as accurate or insightful accounts of who they are".
  - **Predictive algorithms restrict your choices**, e.g. predictive text may lead you to bland non-emotional writing.
  - **Identity feedback loops** (e.g. self-reinforcing recommendations of content and connections) inhibit personal evolution.
  - An **AI's emotional intelligence, based on learned patterns**, may promote **emotional uniformity** rather then emotional intelligence.

*Reference:* Jeena Joseph, "The algorithmic self: how AI is reshaping human identity, introspection, and agency", 2025. https://doi.org/10.3389/fpsyg.2025.1645795

# What to do?

Users and schools:

- Develop “digital habits that prioritize reflective awareness, diverse media consumption, and the scrutiny of AI recommendations”.

Information system designers and developers:

- Allow users to compose personal interpretations of data.
- Provide time for reflection.
- Include counterarguments to interrupt preference loops.
- Reduce data retention to enable *social forgetfulness.*
    - Allow users to autonomously constructing and reconstructing their self-identity without being stuck with former personal and social identities.
    - This must be balanced with retaining accountability where important.
- Avoid *normalization* where “The system watches what you do; it fits you into a pattern; the pattern is then fed back to you in the form of options set by the pattern; the options reinforce the patterns; the cycle begins again” (Lessig)

*References*

Jeena Joseph, “The algorithmic self: how AI is reshaping human identity, introspection, and agency”, 2025. https://doi.org/10.3389/fpsyg.2025.1645795

Blanchette and Johnson, “Data Retention and the Panoptic Society: The Social Benefits of Forgetfulness”, 2011. https://doi.org/10.1080/01972240252818216

Noëmi Manders-Huits, “Practical versus moral identities in identity management”, 2010. https://link.springer.com/article/10.1007/s10676-010-9216-8

Lawrence Lessig, Code and Other Laws of Cyberspace, Basic Books, 1999

# Computing and culture

# Guiding questions on Cultural Diversity

"Applications of emerging ICTs function on a global scale, across national and cultural borders. This raises concerns about dealing with and respecting cultural differences and doing justice to and cultivating cultural diversity. Conceptions and valuations of privacy, for instance, vary significantly across cultures, making it difficult to establish unified policies protecting privacy."

Guiding questions

- Which assumptions about normal and desirable behaviour are embedded in the technology? (We will broaden this to wider assumptions about data collection, management and use)
- Is there a possibility of testing the technology in different cultures?

*Reference:* Stahl, Timmermans and Flick, "Ethics of Emerging Information and Communication Technologies: On the implementation of responsible research and innovation", 2017. https://doi.org/10.1093/scipol/scw069

# Computing and culture

## 1) Data collection, storage and use

# Western vs. indigenous knowledge

| Dimension | Dominant Western scientific tradition | Many indigenous knowledge traditions |
|---|---|---|
| **Ontology** (the nature of things) | World understood as comprising phenomena that can be studied as objects; strong distinction between human observer and natural world | Humans are often understood as part of a network of relationships involving other humans, animals, plants, land, water, ancestors and spiritual entities |
| **Epistemology** (What and how we can know) | Knowledge generated through systematic observation, measurement, hypothesis, experimentation, modelling and inference | Knowledge developed through long-term observation, experience, participation, oral transmission, practice, relationships and intergenerational experience |
| **Role of observer** | Ideally detached or objective observer | Situated participant whose relationship with the subject of knowledge matters |
| **Knowledge ownership** | Tends toward publication, dissemination and incorporation into a collective scientific literature | May be *collectively held*, with different people/groups having different rights and responsibilities concerning particular knowledge |
| **Access to knowledge** | Normatively favours openness and reproducibility | Access may be restricted according to cultural, familial, gender, ceremonial, territorial or other responsibilities |
| **Data** | Generally treated as representations/evidence that can be collected, stored, analysed and reused | May be understood as embedded in relationships, people, places and obligations, so extraction and reuse can require permission and reciprocity |

# A common scenario (at least in the past)

1. A researcher enters an Indigenous community.
2. The community possesses knowledge of interest to the researcher, e.g., about a plant, ecosystem, environment or cultural practice.
3. The researcher records that knowledge as observations.
4. The observations enter a database as data.
5. The database makes them accessible to other researchers.
6. Someone develops a commercial application.
7. The original knowledge holders have little control over the subsequent uses and gain no benefit.

Is this

*knowledge dissemination and scientific progress*

or

*knowledge extraction*?

# And today it is increasingly automated

From a report of an interview with Maori AI researcher Te Taka Keegan (2023):

- We're staring down the barrel of "digital colonisation".
- Chat GPT is already speaking te reo Māori, composing waiata Māori and even writing karakia. It's doing that by taking data it shouldn't necessarily have access to.

Source: https://e-tangata.co.nz/comment-and-analysis/the-dangers-of-digital-colonisation/

# Data management and use across three perspectives

| | Western scientific tradition | Western bureaucratic/government tradition | Indigenous / relational perspective |
|---|---|---|---|
| Fundamental question | What can we know? | What do we need to know to govern? | What do we know, through what relationships, and what responsibilities follow? |
| Data represents | Observations about the world | People, populations and phenomena in administratively useful categories | Knowledge embedded in people, places, relationships and obligations |
| Collection | Observe and measure systematically | Collect information needed for administration and policy | Collect with appropriate authority, purpose and relationships |
| Storage | Preserve knowledge for future research | Maintain institutional records and organisational memory | Retain knowledge under appropriate guardianship and control |
| Sharing | Openness facilitates scientific progress | Sharing enables coordination and effective government | Sharing depends on authority, purpose, relationships and benefit |
| Aggregation | Enables generalisation | Makes populations governable | Can erase context, identity and relationships |
| Purpose | Understand, explain, predict | Administer, regulate, plan, allocate | Sustain relationships, wellbeing and collective interests |
| Key value | Validity | Utility | Responsibility / relationships |

# CARE Principles for Indigenous Data Governance

Research data should be used to support indigenous communities, not just benefit researchers.

Data use must minimize harm and maximize benefits. Indigenous Peoples' rights and wellbeing should be considered at all stages of the data life cycle.




Indigenous people have the right to control the data that relates to them, including lands and culture.

Those working with indigenous data must treat it with respect and share how the data can support indigenous people's self-determination and be of benefit to them.

Diagram by by Connie Strayer and Robyn Grebliunas. Source: https://opentextbc.ca/indigenousdigitalliteracies/chapter/urban-indigenous-governance/. Licence CC BY 4.0. Summaries from https://aut.ac.nz.libguides.com/c.php?g=982265&p=7157663.

# NZ context: The Māori Data Sovereignty Principles

**Kotahitanga | Collective benefit**

- Data is used to benefit the people it is about
- Researchers and organisations should actively support Māori communities in developing the skills and knowledge necessary to manage and use data effectively.
- Connections between Māori and other Indigenous peoples shall be supported to enable the sharing of strategies, resources and ideas in relation to data.

**Whanaungatanga | Obligations**

- Individual rights need to be balance with collective rights dependent on context.
- Building relationships within the community and the obligations that come with these relationships.

**Kaitiakitanga | Guardianship**

- Tikanga (procedure/custom), kawa (protocols) and mātauranga (knowledge) underpin the access and use of data.

**Manaakitanga | Reciprocity**

- Free, prior and informed consent shall underpin the collection of data.




**Rangatiratanga | Authority**

- Recognises the rights that people have over their data, including how it is stored, managed, used and shared.
- Whenever possible, Māori data shall be stored in Aotearoa New Zealand.
- Projects that focus on Māori as a key population should be led or co-led by Māori Principal Investigators.

**Kaitiakitanga | Guardianship**

- Data is stored in a way that Māori can still exercise kaitiakitanga over it.
- Māori decide which is tapu or noa about their own data.

**Whakapapa | Relationships**

- The origins and context of data is of great importance.
- Disaggregation/ structuring of data to be more relevant at iwi or hāpu level.
- A key goal of Māori data governance should be to protect against future harm.

**Manaakitanga | Reciprocity**

- 'Mana enhancing'. Ensuring that data is used respectfully, to uplift people.
- Free, prior and informed consent shall underpin the collection of data.

Based on https://aut.ac.nz.libguides.com/c.php?g=982265&p=7157663.
For a full version of the Māori Data Sovereignty Principles, see https://www.temanararaunga.maori.nz/s/TMR-Maori-Data-Sovereignty-Principles-Oct-2018.pdf

# What to do?

- If you collect, store, manage, process, share, publish and/or commercialise data in New Zealand, you should be aware of the Māori Data Sovereignty Principles and respect them when working with *Maori data*.
- "Māori data refers to digital or digitizable information or knowledge that is about or from Māori people, our language, culture, resources or environments."
  https://www.temanararaunga.maori.nz/s/TMR-Maori-Data-Sovereignty-Principles-Oct-2018.pdf

# Computing and culture

## 2) AI and cultural bias

# Cultural bias in AI

- AI models are trained using huge amounts of data harvested from the Internet and public domain  books. This is dominated by material in English and by Western authors.
- The AI workforce lacks diversity (see the demographics in Lecture 2).
- These lead to biased models and a lack of awareness of the effects of that bias.
- Example of bias:
  - In 2023, Internet media company Buzzfeed published an article featuring AI-generated articles of what "Barbie would look like in every country in the world" Cultural offensive or biased representations included:

> **German Barbie's** costume looked like a Nazi SS uniform. **South Sudan Barbie** was holding a gun. **Vietnamese Barbie's** collar was reversed, symbolising death. A few **Middle Eastern Barbies** wore a man's headdress and the **Tajikistan and Thai Barbies** had blonde hair.
> Source: https://www.businessinsider.com/ai-generated-barbie-every-country-criticism-internet-midjourney-racism-2023-7

Original Buzzfeed article (warning, this may offend): https://web.archive.org/web/20230708232040/https://www.buzzfeed.com/aglover/from-afghanistan-to-zimbabwe-heres-what-barbie-would-look?d_id=6036297&ref=bftwbuzzfeed

# Low-resource languages and Large Language Models (LLMs)

- *A low-resource language* is one that has limited digital datasets available, reducing the effectiveness of training on that language.
- This has social impacts on speakers of that language as AI use becomes the "go to" tool for many tasks.
- More from the interview with Te Kaka Keegan (slide 16):
  - … we'll become too reliant on Chat GPT for reo growth and development. Every time we need a new kupu, phrase, waiata or karakia, we'll turn to it as the easiest, most convenient option, rather than our original sources.
  - We'll no longer have different dialects or learn kupu from different iwi. Our reo will become one homogenised language.
- Text in a culture's language conveys narratives about the culture. Limited training data "risks flattening rich, diverse cultures into oversimplified narratives that fail to capture their true complexity" with the AI defaulting to Western narratives.

https://medium.com/@nardosdemilew/ai-is-killing-languages-why-low-resource-languages-deserve-more-attention-in-ai-development-0f55f108bd9f

# What to do?

- Set up teams of subject experts and developers to identify possible sources of bias *in advance* (for all source of bias, not just cultural!)
- Use various technical methods to augment datasets and give minority classes of data more weight in training.
- Analyse the trained model's bias using various toolkits for developers.
- National and cultural groups can curate their own datasets and/or build their own models or partner with Big Tech:
  - In 2013, Google added Te Reo Māori to Google Translate after partnering with Te Taka Keegan and a volunteer group of volunteer English to te Reo Māori translators. (However, it is still not always accurate).
  - In 2022, Straker, Microsoft and Stuff formed a partnership to build an English to te reo Māori translator to increase the number of bilingual articles on stuff.co.nz. Stuff started using it in 2024 (with quality checks from a human translator).
- Develop national 'sovereign AI' large language models such as FinGPT-3 and Switzerland's Aspertus.

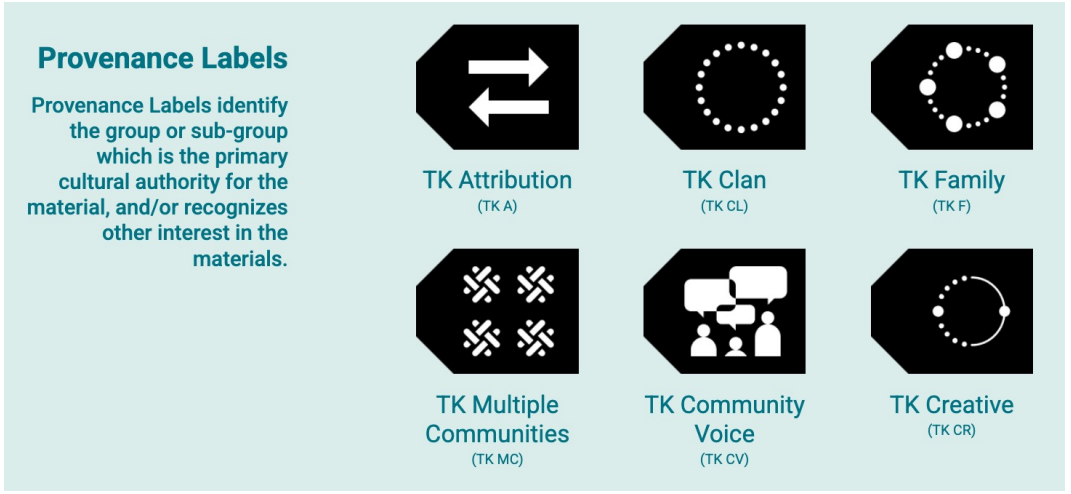
# Computing and culture

## 3) Using IT to preserve and enrich culture

# Covered in Lab 7

- Traditional Knowledge (TK) and Biocultural (BC) labels
  - Address a principle from **The principles of Māori data sovereignty**:

    **2.1 Context**. All data has a whakapapa (genealogy). Accurate metadata should, at minimum, provide information about the provenance of the data, the purpose(s) for its collection, the context of its collection, and the parties involved."







Examples from https://localcontexts.org/labels/traditional-knowledge-labels/

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# Covered in Lab 7

- The Ngai Tahu Atlas: preserving place-based Māori knowledge using a Geographic information system (GIS)

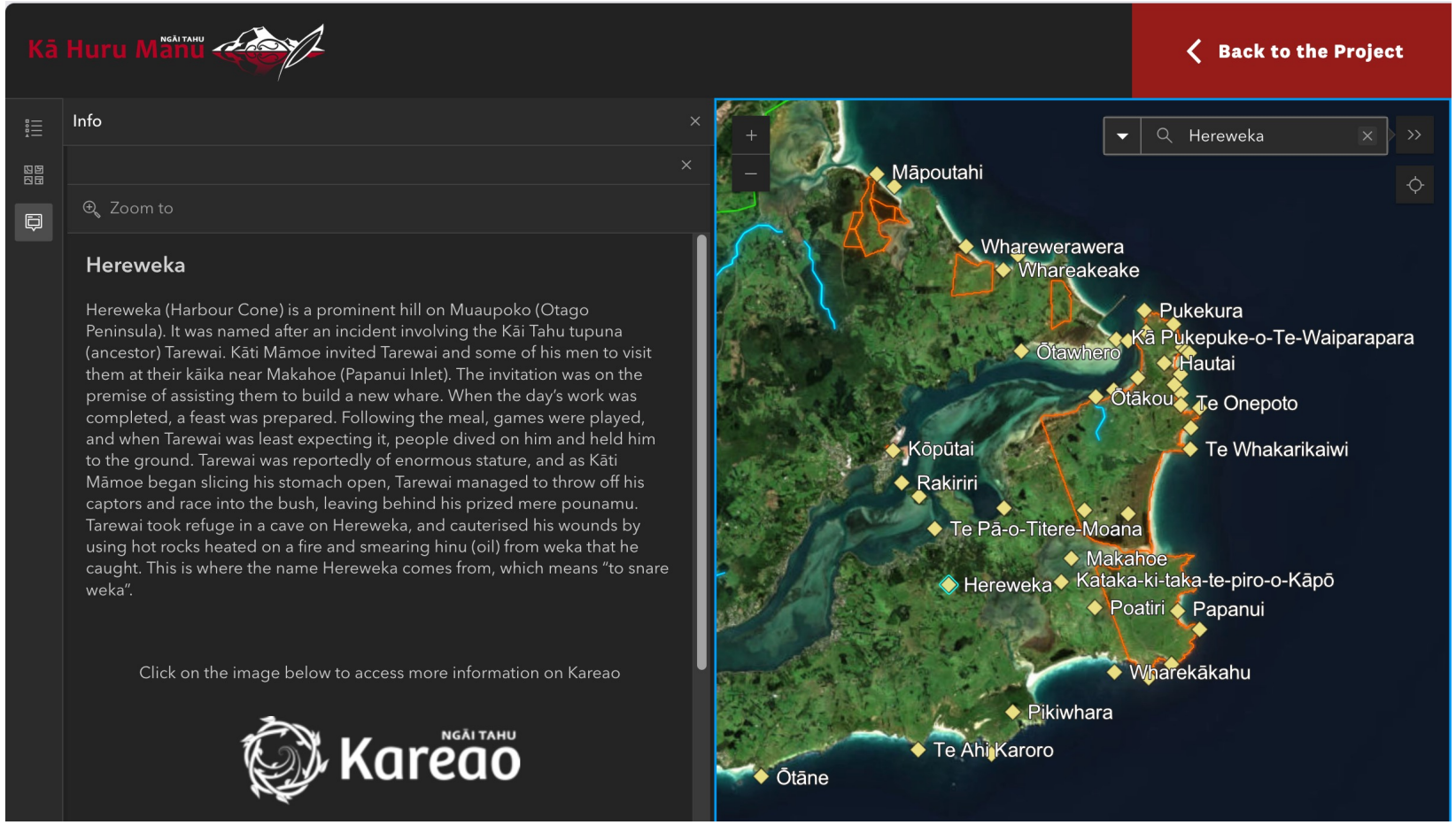



Image from https://kahurumanu.co.nz/atlas